

1. (10%) Order the following function by growth rate in increasing order:

n^2 , $n \log(\log n)$, $n \log(n^2)$, n^n

$$n \log n^2 = 2n \log n, \quad 2n \log n > n \log(\log n)$$

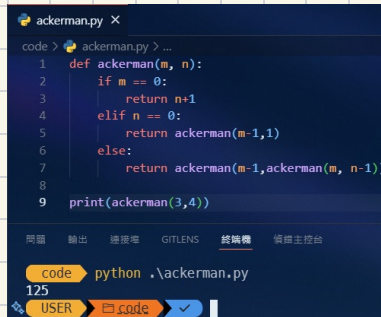
$$\text{Ans: } n \log(\log n) < n \log(n^2) < n^2 < n^n \quad \#$$

2. (10%) Ackerman's function $A(m, n)$ is defined as:

$$A(m, n) = \begin{cases} n + 1, & \text{if } m = 0 \\ A(m - 1, 1), & \text{if } n = 0 \\ A(m - 1, A(m, n - 1)), & \text{otherwise} \end{cases}$$

What is the value of $A(3, 4)$?

$$\text{Ans: } 125 \quad \#$$



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ackerman.py X
code > ackerman.py > ...
1 def ackerman(m, n):
2     if m == 0:
3         return n+1
4     elif n == 0:
5         return ackerman(m-1, 1)
6     else:
7         return ackerman(m-1, ackerman(m, n-1))
8
9 print(ackerman(3, 4))

125
USER
```

3. (12%) Show that the following statements are correct:

(a) $5n^2 - 6n = \Theta(n^2)$

(b) $n! = O(n^n)$

(a)

$$\text{let } f(n) = 5n^2 - 6n, \quad g(n) = n^2$$

$$\text{take } c_1 = 2, \quad c_2 = 6, \quad \text{for all } n \geq 2, \quad c_1 g(n) \leq f(n) \leq c_2 g(n)$$

$$\therefore 5n^2 - 6n = \Theta(n^2) \quad \#$$

(b)

$$\text{let } f(n) = n!, \quad g(n) = n^n$$

$$\text{take } c = 1, \quad \text{for all } n \geq 1, \quad f(n) \leq c g(n) \quad (n! = n \times (n-1) \times \dots \times 2 \times 1 \leq n^n = \underbrace{n \times n \times \dots \times n}_{n \text{ 個 } n})$$

$$\therefore n! = O(n^n) \quad \#$$

4. (12%) Show that the following statements are incorrect:

(a) $10n^2 + 9 = O(n)$

(b) $3^n = O(2^n)$

(a)

let $f(n) = 10n^2 + 9$, $g(n) = n$

若 $10n^2 + 9 = O(n)$, 則表示存在一個常數 C 和 n_0 , 使得對於所有 $n \geq n_0$, $f(n) \leq Cg(n)$ 成立, 即 $10n^2 + 9 \leq cn$

$10n^2 + 9 \leq cn$ 同除 n , 可得 $10n + \frac{9}{n} \leq C$

$\lim_{n \rightarrow \infty} 10n + \frac{9}{n} = \infty$, 而一常數必不可能大於可能趨近無限大的函數值, 故得證。

(b)

let $f(n) = 3^n$, $g(n) = 2^n$

若 $3^n = O(2^n)$, 則表示存在一個常數 C 和 n_0 , 使得對於所有 $n \geq n_0$, $f(n) \leq Cg(n)$ 成立, 即 $3^n \leq C2^n$

$3^n \leq C2^n$ 同除 2^n , 可得 $(\frac{3}{2})^n \leq C$

$\lim_{n \rightarrow \infty} (\frac{3}{2})^n = \infty$, 而一常數必不可能大於可能趨近無限大的函數值, 故得證。

5. (16%) (from NTHU 106) Which ones of the following conjectures are correct?

Please justify your answers:

~~(a)~~ $3n^2 - 100n + 6 = O(n)$

(b) $\log(n!) = \Theta(n \log n)$

(c) $n! = O(n^n)$

(d) $(n+a)^b = \Theta(n^b)$ where a and b are real constants

~~(e)~~ $(n+a)^b = \Omega(n^{b+1})$ where a and b are real constants

(a)

$$3n^2 - 100n + 6 \neq O(n)$$

因為不管常數為何， n^2 增長的速度一定比 n 快。

(b)

$$\text{let } f(n) = \log(n!), g(n) = n \log n$$

$$\text{take } c_1 = \frac{1}{2}, c_2 = 4, \text{ for all } n \geq 1, c_1 g(n) \leq f(n) \leq c_2 g(n)$$

$$\therefore \log(n!) = \Theta(n \log n)$$

(c)

$$\text{let } f(n) = n!, g(n) = n^n$$

$$\text{take } c = 1, \text{ for all } n \geq 1, f(n) \leq c g(n), \therefore n! = O(n^n)$$

(d)

$$\lim_{n \rightarrow \infty} \frac{(n+a)^b}{n^b} = \left(\frac{n+a}{n}\right)^b = \left(1 + \frac{a}{n}\right)^b = 1$$

故存在 $C_1, C_2 > 0$ 和 n_0 , s.t. 所有 $n \geq n_0$ 有 $C_1 n^b \leq (n+a)^b \leq C_2 n^b$

$$\therefore (n+a)^b = \Theta(n^b)$$

(e)

$$\lim_{n \rightarrow \infty} \frac{(n+a)^b}{n^{b+1}} = \left(\frac{n+a}{n}\right)^b \times \frac{1}{n} = \frac{a}{n^2} + \frac{1}{n} = 0$$

由於極限值為 0，代表 $(n+a)^b$ 遠小於 n^{b+1} ，因此 $(n+a)^b \neq \Omega(n^{b+1})$

Ans: (b), (c), (d)

6. (20%) Fill the step count table (yellow regions) for the following program:

Statement	s/e	Frequency	Total Steps
* Let $MAX_SIZE = n$			
void mult(int a[][MAX_SIZE],int b[][MAX_SIZE], int c[][MAX_SIZE])	0	0	0
{	0	0	0
int i,j,k;	0	0	0
for (i = 0; i < MAX_SIZE; i++)	1	$n+1$	$n+1$
for (j = 0; j < MAX_SIZE; j++) {	1	$n \times (n+1)$	n^2+n
c [i][j] = 0;	1	n^2	n^2
for (k = 0; k < MAX_SIZE; k++)	1	$n^2 \times (n+1)$	n^3+n^2
c [i][j] += a [i][k] * b [k][j];	1	n^3	n^3
}	0	0	0
}	0	0	0
Total			$2n^3+3n^2+2n+1$

7. (10%) Given an four-dimensional array A[100][200][300][400], if the address of A[3][20][32][45] is $\alpha + X$, what is the value of X ?

Note: (1) α is the address of A[0][0][0][0]

(2) Only consider row major here.

$$\alpha + 3 \times 200 \times 300 \times 400 + 20 \times 300 \times 400 + 32 \times 400 + 45$$

$$= \alpha + 72000000 + 2400000 + 12800 + 45 = 74412845$$

8. (10%) (from NTCU 107) Consider the following function. Please derive the time complexity of function foo using recurrence relation. Please show your derivation step by step.

	S / e	Frequency	Total	Steps
int i[n];	0	0	0	
int foo(int a, int b, int c) {	0	0	0	
int d;	0	0	0	
if (a>b) return -1;	1	t	t	t
d=(a+b)/2;	1	t	t	t
if (c==i[d])	1	t	t	t
return d;	1	1	1	1
else if (c>i[d])	1	t	t	t
return foo(d+1,b,c);	1	t	t	t
else	1	t	t	t
return foo(a, d-1,c);	1	t	t	t
}				6t + 1

Assume the function foo runs t times.

最差的情況下，function foo 會不斷透過 a、b 將陣列 n 拆半，直到 $n=2^t$ 時 $n=2^t$ ， $t=\log_2 n$ ，因此，此程式的 complexity 是 $6\log_2 n + 1 = O(\log n)$